

An Analysis of Time series forecasting

Time series forecasting

Time series forecasting -- Part 1

Estimation, filtering, and smoothing This approach may be based on harmonic analysis and filtering of signals in the frequency domain using the Fourier transform, and spectral density estimation. Its development was significantly accelerated during World War II by mathematician Norbert Wiener, electrical engineers Rudolf E. Kálmán, Dennis Gabor and others for filtering signals from noise and predicting signal values at a certain point in time. An equivalent effect may be achieved in the time domain, as in a Kalman filter; see filtering and smoothing for more techniques. Other related techniques include:

Time series forecasting -- Part 2

Motivation In the context of statistics, econometrics, quantitative finance, seismology, meteorology, and geophysics the primary goal of time series analysis is forecasting. In the context of signal processing, control engineering and communication engineering it is used for signal detection. Other applications are in data mining, pattern recognition and machine learning, where time series analysis can be used for clustering, classification, query by content, anomaly detection as well as forecasting.

Time series forecasting -- Part 3

Stationary process Ergodic process Ergodicity implies stationarity, but the converse is not necessarily the case. Stationarity is usually classified into strict stationarity and wide-sense or second-order stationarity. Both models and applications can be developed under each of these conditions, although the models in the latter case might be considered as only partly specified. In addition, time-series analysis can be applied where the series are seasonally stationary or non-stationary. Situations where the amplitudes of frequency components change with time can be dealt with in time-frequency analysis which makes use of a time–frequency representation of a time-series or signal.

Time series forecasting -- Part 4

Methods for analysis Methods for time series analysis may be divided into two classes: frequency-domain methods and time-domain methods. The former include spectral analysis and wavelet analysis; the latter include auto-correlation and cross-correlation analysis. In the time domain, correlation and analysis can be made in a filter-like manner using scaled correlation, thereby mitigating the need to operate in the frequency domain.

Time series forecasting -- Part 5

Autocorrelation analysis to examine serial dependence Spectral analysis to examine cyclic behavior which need not be related to seasonality. For example, sunspot activity varies over 11 year cycles. Other common examples include celestial phenomena, weather patterns, neural activity, commodity prices, and economic activity. Separation into components representing trend, seasonality, slow and fast variation, and cyclical irregularity: see trend estimation and decomposition of time series

Time series forecasting -- Part 6

Prediction and forecasting In statistics, prediction is a part of statistical inference. One particular approach to such inference is known as predictive inference, but the prediction can be undertaken within any of the several approaches to statistical inference. Indeed, one description of statistics is that it provides a means of transferring knowledge about a sample of a population to the whole population, and to other related populations, which is not necessarily the same as prediction over time. When information is transferred across time, often to specific points in time, the process is known as forecasting.

Time series forecasting -- Part 7

Subsequence time series clustering Subsequence time series clustering resulted in unstable (random) clusters induced by the feature extraction using chunking with sliding windows. It was found that the cluster centers (the average of the time series in a cluster - also a time series) follow an arbitrarily shifted sine pattern (regardless of the dataset, even on realizations of a random walk). This means that the found cluster centers are non-descriptive for the dataset because the cluster centers are always nonrepresentative sine waves.

Time series forecasting -- Part 8

Consideration of the autocorrelation function and the spectral density function (also cross-correlation functions and cross-spectral density functions) Scaled cross- and auto-correlation functions to remove contributions of slow components Performing a Fourier transform to investigate the series in the frequency domain Performing a clustering analysis Discrete, continuous or mixed spectra of time series, depending on whether the time series contains a (generalized) harmonic signal or not Use of a filter to remove unwanted noise Principal component analysis (or empirical orthogonal function analysis) Singular spectrum analysis "Structural" models: General state space models Unobserved components models Machine learning Artificial neural networks Support vector machine Fuzzy logic Gaussian process Genetic programming Gene expression

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programming Hidden Markov model Multi expression programming Queueing theory analysis Control chart Shewhart individuals control chart CUSUM chart EWMA chart Detrended fluctuation analysis Nonlinear mixed-effects modeling Dynamic time warping Dynamic Bayesian network Time-frequency analysis techniques: Fast Fourier transform Continuous wavelet transform Short-time Fourier transform Chirplet transform Fractional Fourier transform Chaotic analysis Correlation dimension Recurrence plots Recurrence quantification analysis Lyapunov exponents Entropy encoding

heteroskedasticity (ARCH) and the collection comprises a wide variety of representation (GARCH, TARCH, EGARCH, FIGARCH, CGARCH, etc.). Here changes in variability are related to, or predicted by, recent past values of the observed series. This is in contrast to other possible representations of locally varying variability, where the variability might be modeled as being driven by a separate time-varying process, as in a doubly stochastic model.

Time series forecasting -- Part 9

Other modeling approaches In a recent work on "model-free" analyses (a term often used to refer to analyses that do not rely on modeling the processes evolution over time with a parametric mathematical expression), wavelet transform based methods (for example locally stationary wavelets and wavelet decomposed neural networks) have gained favor. Multiscale (often referred to as multiresolution) techniques decompose a given time series, attempting to illustrate time dependence at multiple scales. See also Markov switching multifractal (MSMF) techniques for modeling volatility evolution. A hidden Markov model (HMM) is a statistical Markov model in which the system being modeled is assumed to be a Markov process with unobserved (hidden) states. An HMM can be considered as the simplest dynamic Bayesian network. HMM models are widely used in speech recognition, for translating a time series of spoken words into text. Many of these models are collected in the python package `sktime`.

Time series forecasting -- Part 10

History Jevons was the first to study times series. In his book he examined a number of weekly financial time series between the years 1825 to 1860 inclusive. These included times series of bankruptcies, currency circulation and discount rates.

Time series forecasting -- Part 11

Non-linear models Non-linear dependence of the level of a series on previous data points is of interest, partly because of the possibility of producing a chaotic time series. However, more importantly, empirical investigations can indicate the advantage of using predictions derived from non-linear models, over those from linear models, as for example in nonlinear autoregressive exogenous models. Further references on nonlinear time series analysis: (Kantz and Schreiber), and (Abarbanel) Among other types of non-linear time series models, there are models to represent the changes of variance over time (heteroskedasticity). These models represent autoregressive conditional